

DS	Session	Hashing	Question Information	Level 1	Challenge 98
Question description You are given an array A of length N which is initialised with 0. You will be given Q queries of two types: 1 k: set value 1 at index k in array A 2 y: print the smallest index x which is greater than or equal to y and having value 1. If there is no such index print -1. Note: Indexing is 1 based Constraints $1 \leq n \leq 10^9$ $1 \leq q \leq 5 * 10^5$ $1 \leq y, k \leq n$ Input Format First line contains two integers N and Q separated by a space. The next Q lines contain the type of query (i.e. either a 1 or a 2), then a space, then for type 1 queries integer k and for type 2 queries integer y Output Format For each query type 2, print in new line, the smallest index x which is greater than or equal to y and having value 1. If there is no such index print -1. Explanation for Test case 1 For first query: $2\ 3$, there is no index greater than or equal index 3, having value -1, so the answer is -1. For second query: $1\ 2$, set value -1 at index 2. For third query: $2\ 1$, index 2 is greater than index 1, having value -1, so the answer is 2. For fourth query: $2\ 3$, there is no index greater than or equal index 3, having value -1, so the answer is -1. For fifth query: $2\ 2$, index 2 is equal to index 2, having value -1, so the answer is 2.					

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 5
2 3
1 2
2 1
2 3
2 2
```

EXPECTED OUTPUT

```
-1
2
-1
2
```

Test Case 2

INPUT (STDIN)

```
5 5
2 1
1 2
2 2
1 2
2 3
```

EXPECTED OUTPUT

```
-1
2
-1
```

✓ Mandatory Test Cases

Test Case 1

KEYWORD

```
for(i=0;i<q;i++)
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

225

Test Case 3

NLOC

38